

CSE 211 Digital Design

Akdeniz University

Week01: Introduction to Digital Design

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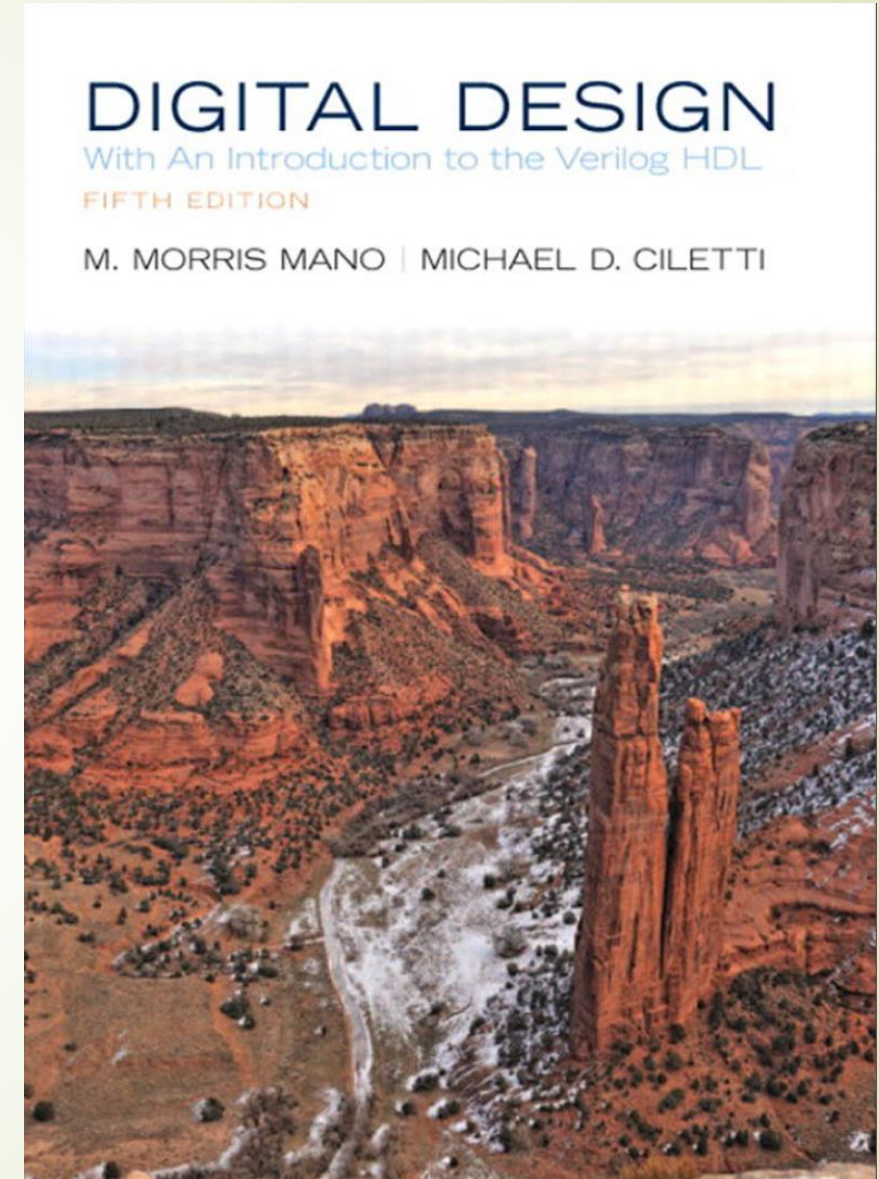
Assoc.Prof.Dr. Taner Danişman

tdanisman@akdeniz.edu.tr

Fall Semester		Spring Semester
August 15, 2025	Last Day for Reporting Course Assignments to the Rectorate	January 2, 2026
August 22, 2025	Last Day for Special Student Applications	January 16, 2026
September 8-12, 2025	Tuition Fee Payment and Registration Renewal Deadline	January 26-30, 2026
September 12, 2025	Last Day to Apply for Study Leave	January 30, 2026
September 15, 2025	Classes Begin	February 2, 2026
September 15, 2025	Last Day to Notify the Rectorate of Courses to be Closed or Allocated to Branches	February 2, 2026
September 16-19, 2025	Drop and Add Period (Add-Drop)	February 3-6, 2026
October 3, 2025	Last Day to Withdraw from a Course	February 20, 2026
December 21, 2025	Deadline for Automating Midterm Exam Results	May 10, 2026
December 21, 2025	End of Classes	May 10, 2026
December 22, 2025 - January 2, 2026	Final Exams	May 11-22, 2026
January 5, 2026	Last Day for Entering Final Exam Results into the Automation System	June 1, 2026
January 3-10, 2026	Application Dates for the Second End of Year/Semester Exam (Make-up Exam)	May 30 - June 6, 2026
January 12-16, 2026	End of Year/Semester Second Exam (Make-up) Dates	June 8-12, 2026
January 19, 2026	Last Day for Entering Year/Semester End Exam Results into Automation	June 15, 2026

Evaluation (absolute)

- Midterm %30
- Assignments %20
- Final %50
- Teaching Assistant: **Erdoğan TÜRK**
- Academic misconduct: **do not let this happen**
- **Monday 15:30-17:20 (Location : Hukuk Fakültesi Amfi 7)**
- **Wednesday 15:30:17:20 (Location: AMFI 2)**
 - **AMFI 2 will be used for 4 weeks (expected)**
 - **Actual laboratory sessions (Section 1 and Section 2) will begin at the EEM Measurement and Circuit Laboratory**
- **Textbook :** Digital Design, With an Introduction to the Verilog HDL 5th Edition, Morris Mano, Michael Ciletti, Pearson.



Can I use the previous year's Lab grades?

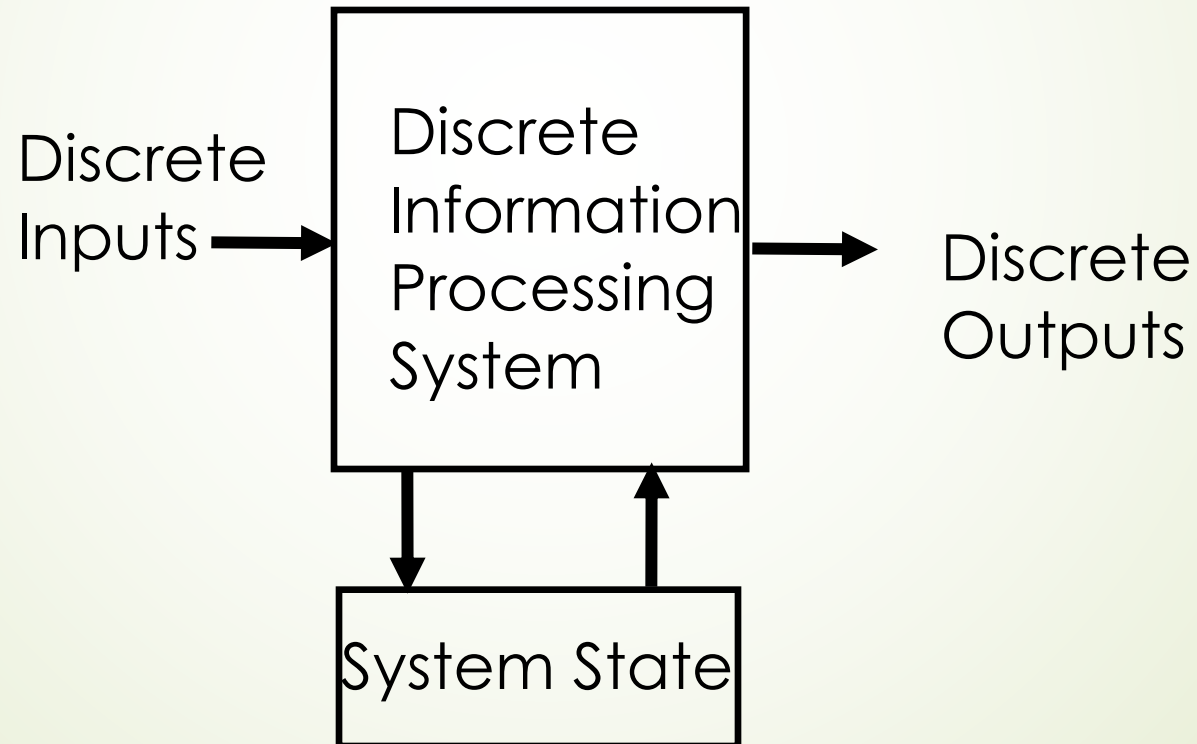
- ➡ **YES**, you can, but you should inform the Teaching Asistant in the beginning of the semestre (first week).

Course program

Week 01	15/09/2025	Introduction
Week 02	22/09/2025	Digital Systems and Binary Numbers I
Week 03	29/09/2025	Digital Systems and Binary Numbers II
Week 04	06/10/2025	Boolean Algebra and Logic Gates I
Week 05	13/10/2025	Boolean Algebra and Logic Gates II
Week 06	20/10/2025	Gate Level Minimization
Week 07	27/10/2025	Karnaugh Maps
Week 08	03/11/2025	Midterm (Expected)
Week 09	10/11/2025	Karnaugh Maps
Week 10	17/11/2025	Combinational Logic
Week 11	24/11/2025	Combinational Logic
Week 12	01/12/2025	Timing, delays and hazards
Week 13	08/12/2025	Synchronous Sequential Logic
Week 14	15/12/2025	Synchronous Sequential Logic

Digital System

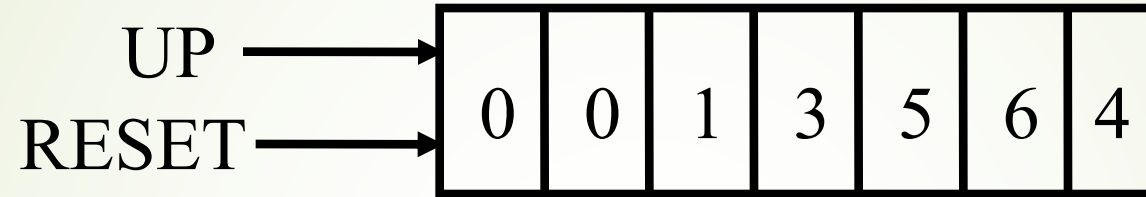
- Takes a set of discrete information inputs and discrete internal information (system state) and generates a set of discrete information outputs.



Types of Systems

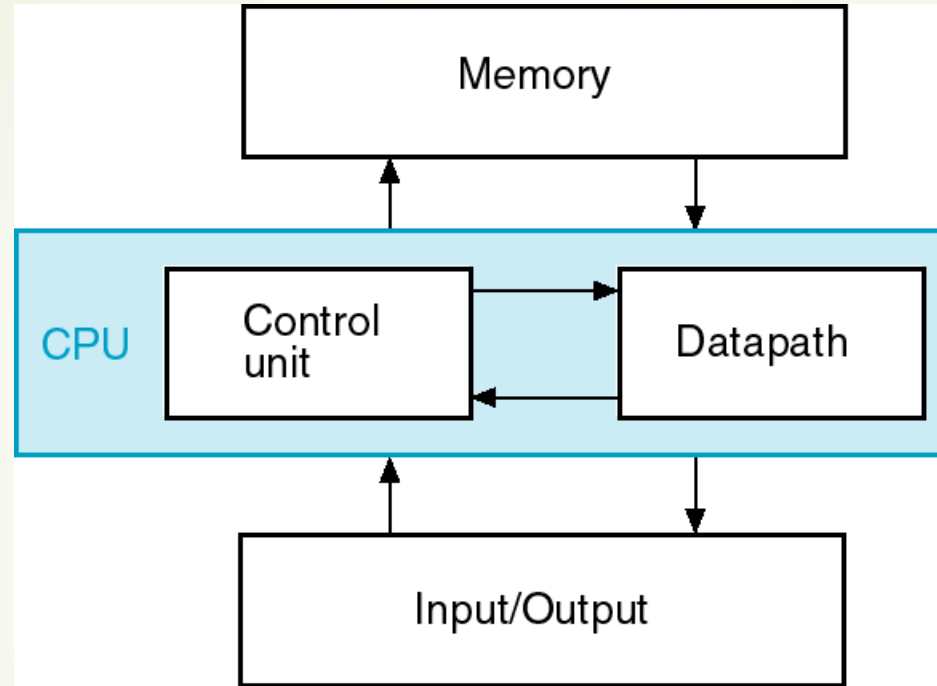
- ▶ With no state present
 - ▶ Combinational logic system
 - ▶ $\text{Output} = \text{Function}(\text{Input})$
- ▶ With state present
 - ▶ State updated at discrete times (e.g., once per clock tick)
 - Synchronous sequential system
 - ▶ State updated at any time
 - Asynchronous sequential system

Example: Digital Counter (e.g., Odometer)



- Inputs: Count Up, Reset
- Outputs: Visual Display
- State: “Value” of stored digits

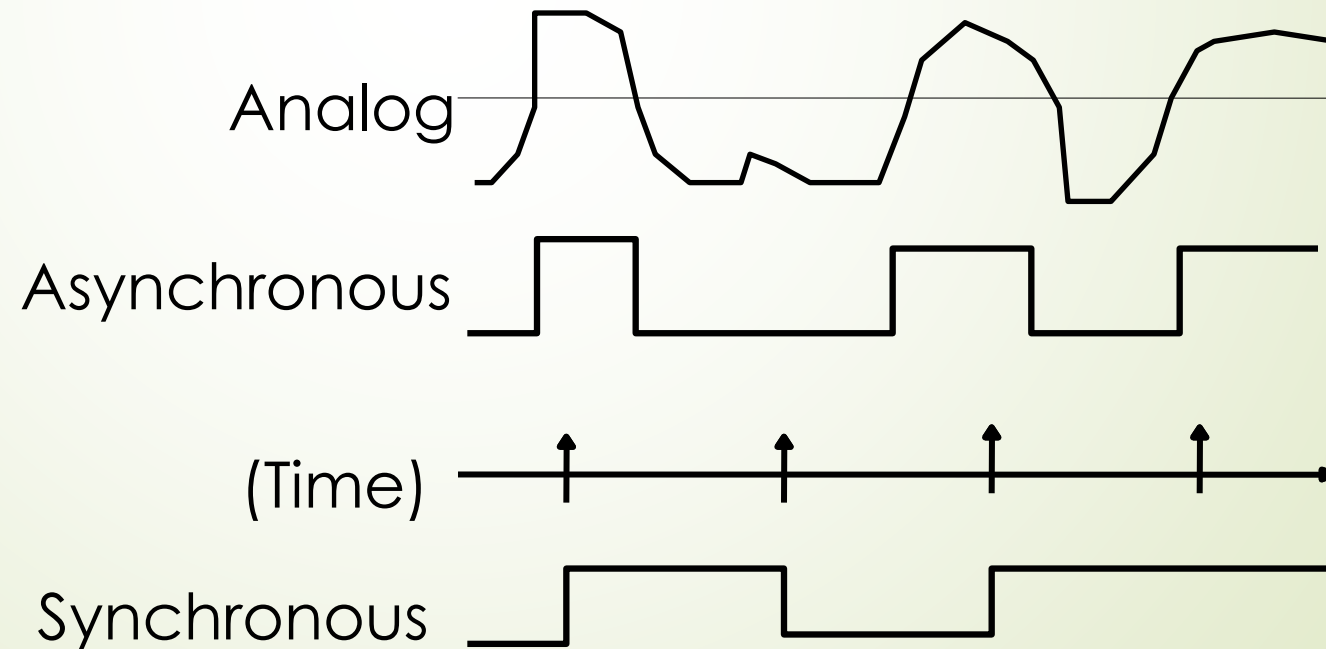
Example: Digital Computer



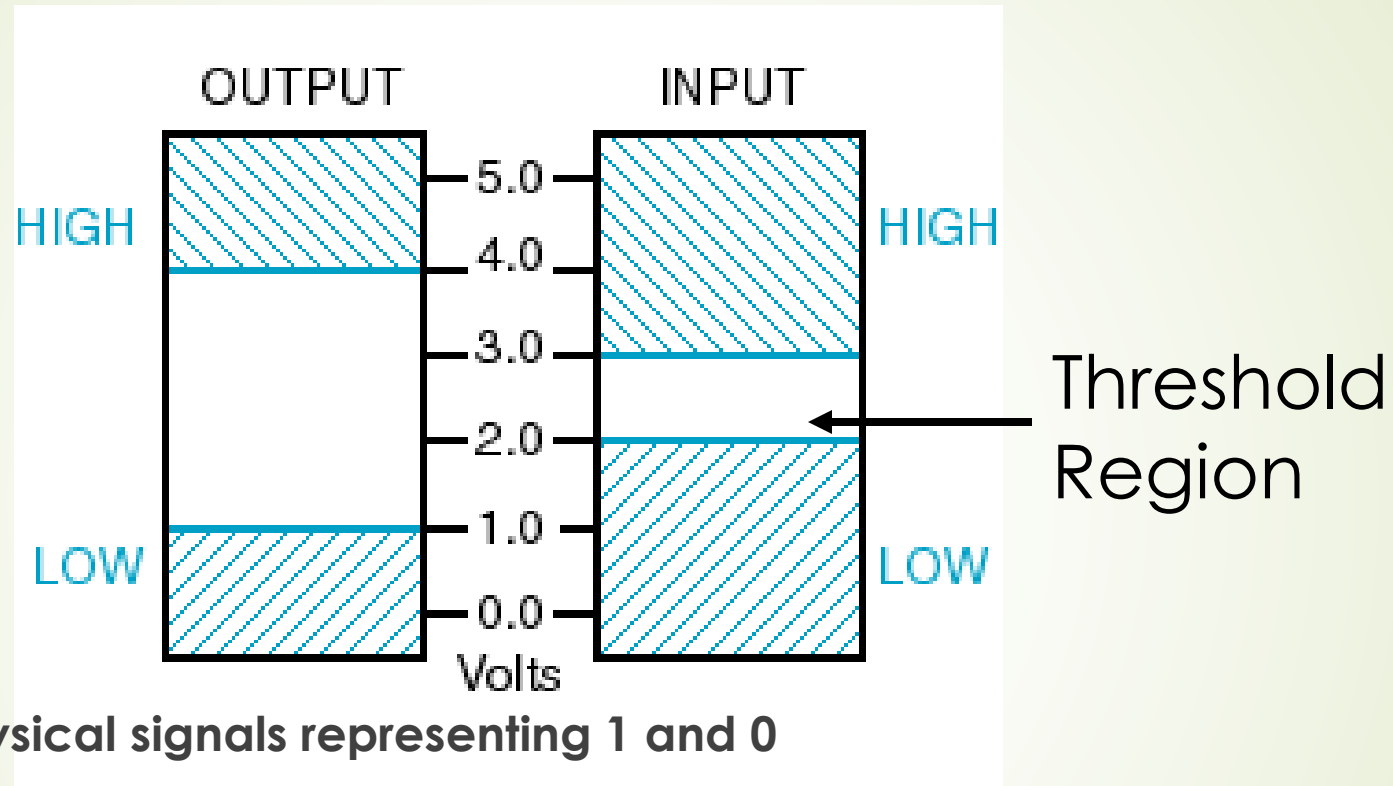
- Inputs: keyboard, mouse, modem, microphone
- Outputs: CRT, LCD, modem, speakers
- Is this system synchronous or asynchronous?

Signals

- Information variables mapped to physical quantities
- In digital systems, the quantities take on *discrete* values
 - Two-level, or *binary*, values are the most prevalent values in digital systems
 - Binary values are represented abstractly by digits 0 and 1
- Signal examples over time:



Physical Signal Example - Voltage



➤ Other physical signals representing 1 and 0

- CPU Voltage
- Disk Magnetic field direction
- CD Surface pits / lands
- Dynamic RAM Charge

Number Systems

➤ Decimal Numbers

➤ What does 5,634 represent?

➤ Expanding 5,634:

$$5 \times 10^3 = 5,000$$

$$+ 6 \times 10^2 = 600$$

$$+ 3 \times 10^1 = 30$$

$$+ 4 \times 10^0 = 4 \rightarrow 5,634$$

➤ What is “10” called in the above expansion?

➤ The radix.

➤ What is this type of number system called?

➤ Decimal.

➤ What are the digits for decimal numbers?

➤ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

➤ What are the digits for radix-r numbers?

➤ 0, 1, ..., r-1.

Powers of 2

- Noteworthy powers of 2:
 - 2^{10} = kilo- = K;
 - 2^{20} = mega- = M;
 - 2^{30} = giga- = G;
 - ... tera-, peta-, ...

General Base Conversion

■ Number Representation

Given a number of radix r of

“ n ” integer digits $a_{n-1}, \dots, a_0$

and

n

“ m ” fractional digits $a_{-1}, \dots, a_{-m}$

written as:

$a_{n-1} a_{n-2} a_{n-3} \dots a_2 a_1 a_0 . a_{-1} a_{-2} \dots a_{-m}$

has value:

$$\begin{aligned} \text{(Number)}_r &= \left(\sum_{i=0}^{i=n-1} a_i \cdot r^i \right) + \left(\sum_{j=-1}^{j=-m} a_j \cdot r^j \right) \\ &\quad \text{(Integer Portion)} + \text{(Fraction Portion)} \end{aligned}$$

Commonly Occurring Bases

Name	Radix	Digits
Binary	2	0,1
Octal	8	0,1,2,3,4,5,6,7
Decimal	10	0,1,2,3,4,5,6,7,8,9
Hexadecimal	16	0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F (= 0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15)

Converting Binary to Decimal

- To convert to decimal, use decimal arithmetic to sum the weighted powers of two:
- Converting 11010_2 to N_{10} :

$$\begin{aligned} N_{10} &= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ &= 26 \end{aligned}$$

Converting Decimal to Binary

- **Method 1 (Method 2 – repeated division – next slide)**
 - Subtract the largest power of 2 that gives a positive result and record the power.
 - Repeat, subtracting from the prior result, until the remainder is zero.
 - Place 1's in the positions in the binary result corresponding to the powers recorded; in all other positions place 0's.
- **Example: $625_{10} \rightarrow 1001110001_2$**
 - $625 - 512 = 113 \rightarrow 9$
 - $113 - 64 = 49 \rightarrow 6$
 - $49 - 32 = 17 \rightarrow 5$
 - $17 - 16 = 1 \rightarrow 4$
 - $1 - 1 = 0 \rightarrow 0$
 - Place 1's in the the positions recorded and 0's elsewhere
- **Converting binary to decimal:** sum weighted powers of 2 using decimal arithmetic, e.g., $512 + 64 + 32 + 16 + 1 = 625$

Conversion Between Bases

➤ Convert the Integral Part

- Repeatedly divide the number by the radix you want to convert to and save the remainders. The new radix digits are the remainders in reverse order of computation.

Why does this work?

This works because, the remainder left in the division is always is the coefficient of the radix's exponent.

- If the new radix is > 10 , then convert all remainders > 10 to digits A, B, ...

➤ Convert the Fractional Part

- Repeatedly multiply the fraction by the radix and save the integer digits that result. The new radix fraction digits are the integer numbers in computed order.

Why does this work?

To convert fractional part, it should be divided by reciprocal of radix, which is same as multiplying with radix.

- If the new radix is > 10 , then convert all integer numbers > 10 to digits A, B, ...

➤ Join together with the radix point

Example: Convert 46.6875_{10} To Base 2

► Convert 46 to Base 2

$$46/2 = 23 \text{ remainder } = 0$$

$$23/2 = 11 \text{ remainder } = 1$$

$$11/2 = 5 \text{ remainder } = 1$$

$$5/2 = 2 \text{ remainder } = 1$$

$$2/2 = 1 \text{ remainder } = 0$$

$$1/2 = 0 \text{ remainder } = 1$$

Read off in reverse order: 101110_2

► Convert 0.6875 to Base 2:

$$0.6875 * 2 = 1.3750 \text{ int} = 1$$

$$0.3750 * 2 = 0.7500 \text{ int} = 0$$

$$0.7500 * 2 = 1.5000 \text{ int} = 1$$

$$0.5000 * 2 = 1.0000 \text{ int} = 1$$

$$0.0000$$

Read off in forward order: 0.1011_2

► Join together with the radix point: 1011110.1011_2

Converting Among Octal, Hexadecimal, Binary

➤ Octal (Hexadecimal) to Binary:

- Restate the octal (hexadecimal) as three (four) binary digits, starting at radix point and going both ways

➤ Binary to Octal (Hexadecimal):

- Group the binary digits into three (four) bit groups starting at the radix point and going both ways, padding with zeros as needed in the fractional part
- Convert each group of three (four) bits to an octal (hexadecimal) digit

➤ Example: Octal to Binary to Hexadecimal

6 3 5 . 1 7 7 ₈

= 110 | 011 | 101 . 001 | 111 | 111 ₂

= 1 | 1001 | 1101 . 0011 | 1111 | 1(000) ₂ (regrouping)

= 1 9 D . 3 F 8 ₁₆ (converting)

Non-numeric Binary Codes

- Given n binary digits (called bits), a binary code is a mapping from a subset of the 2^n binary numbers to some set of represented elements.

- **Example: A binary code for the seven colors of the rainbow**

Binary Number	Color
000	Red
001	Orange
010	Yellow
011	Green
100	(Not mapped)
101	Blue
110	Indigo
111	Violet

- Flexibility of representation
 - can assign binary code word to any numerical or non-numerical data as long as data uniquely encoded.

Number of Bits Required

- Given M elements to be represented by a binary code, the minimum number of bits, n , needed satisfies the following relationships:
 - $2^n \geq M > 2^{n-1}$
 - $n = \text{ceil}(\log_2 M)$ where $\text{ceil}(x)$ is the smallest integer greater than or equal to x
- Example: How many bits are required to represent decimal digits with a binary code?
 - $M = 10 \rightarrow n = 4$

Number of Elements Represented

- Given n digits in radix r , there are r^n distinct elements that can be represented.
- But, can represent m elements, $m < r^n$
- Examples:
 - Can represent 4 elements in radix $r = 2$ with $n = 2$ digits: (00, 01, 10, 11)
 - Can represent 4 elements in radix $r = 2$ with $n = 4$ digits: (0001, 0010, 0100, 1000)
 - This code is called a "one hot" code

Binary Coded Decimal (BCD)

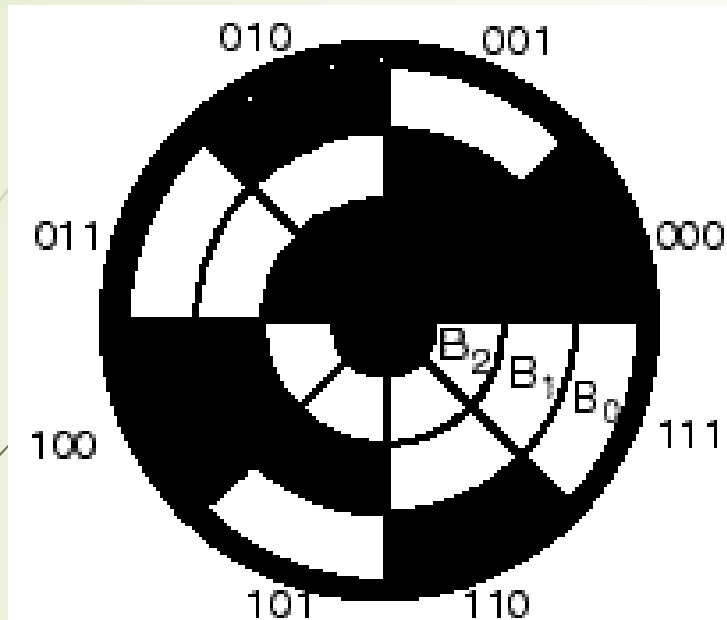
- The BCD code is the 8,4,2,1 code.
- This code is the simplest, most intuitive binary code for decimal digits and uses the same weights as a binary number, but only encodes the first ten values from 0 to 9.
- Example: $1001 (9) = 1000 (8) + 0001 (1)$
- *How many “invalid” code words are there?*
- *What are the “invalid” code words?*

Gray Code

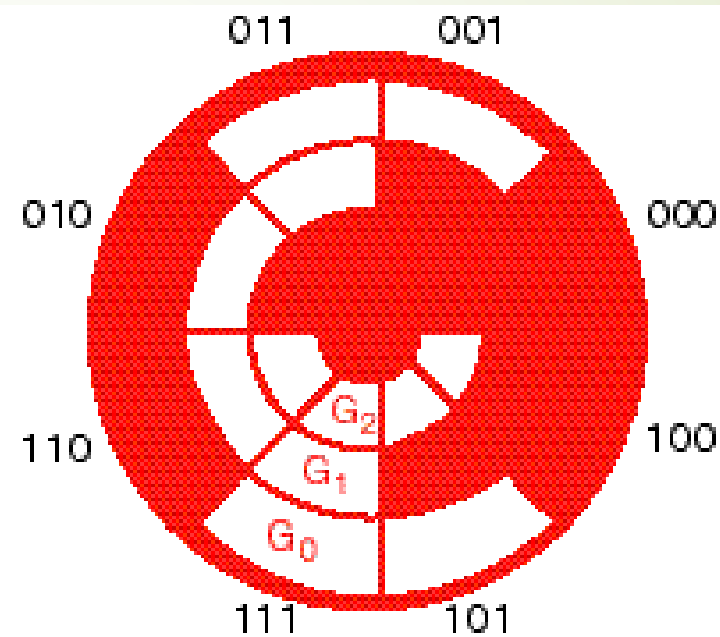
- ▶ **What property does this Gray code have?**
 - ▶ Counting up or down changes only one bit at a time (including counting between 9 and 0)

Decimal	Binary	Gray
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

Gray Code: Optical Shaft Encoder

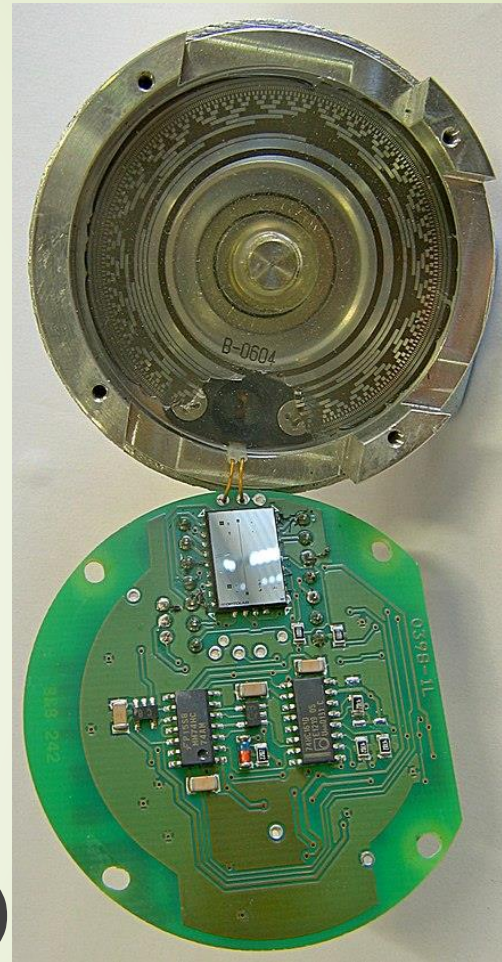


(a) Binary Code for Positions 0 through 7



(b) Gray Code for Positions 0 through 7

- Shaft encoder: Capture angular position (e.g., compass)
- For binary code, what values can be read if the shaft position is at boundary of “3” and “4” (011 and 100) ?
- For Gray code, what values can be read ?



Warning: Conversion or Coding?

- ▶ Do NOT mix up conversion of a decimal number to a binary number with coding a decimal number with a BINARY CODE.
- ▶ $13_{10} = 1101_2$ (This is conversion)
- ▶ $13 \Leftrightarrow 0001 \mid 0011$ (This is coding)